

Initiation à la recherche

Le doctorat

/

Introduction to research

PhD thesis

Module «Projets, recherche et séminaires »

PR422 (Etu) : 2/4 / PR423 (App) : 2/4 ECTS

Bureau : B105 & A203

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Introduction of the speakers

- **Pierre-Alain Toupance**
- PRAG at Esisar since 2010 and researcher at LCIS Laboratory since 2024
- PhD in control theory from Université Grenoble Alpes (UGA) and University of Geneva
- Research activities focused on “Information theory applied to control of complex systems”

Introduction of the speakers

- **Louis Morge-Rollet**
- Associate professor (teacher/researcher) at Esisar and the LCIS Laboratory since 2024
- PhD in telecommunications from ENSTA Bretagne
- Research activities focused on “Analysis of radio frequency signals in embedded context”

Preliminary information

- Objectives of this chapter:
 - Presenting the PhD and the associated steps
 - Presenting the opportunities after the PhD
- This course will also present different PhD journeys from actual and former PhD students

Plan

- What is a PhD ?
- The objectives
- Examples of PhD subjects
- Before PhD thesis
- Steps of the PhD thesis
- Doctoral school
- The advantages of a PhD
- Opportunities
- Testimonials

What is a PhD ?

- PhD: « **Philosophiae Doctor** »
 - The PhD is (one of) the **highest academic degree** in French academic (Bac +8).
 - A personal research project lasting three years
 - Determine a research question (thesis topic).
 - Provide a state-of-the-art review on the subject.
 - Propose innovative solutions to the problem.
 - Develop new products, tools or approaches and rigorously validate them.
 - Promote these results to the scientific community (publications in scientific journals, presentations at conferences, thesis reports, and defense before examiners from the scientific community)

A PhD in a few words

- 1 doctoral student
- 1 thesis topic
- 1 thesis supervisor (+co-supervisor, co-advisors, etc.)
- 1 host organization (research laboratory, company)
- 1 doctoral school
- 1 written report
- 1 defense before a jury
- 1 thesis funding
 - Doctoral contract
 - ANR project, Europe Region
 - CIFRE funding (industrial)
- International experience (« cotutelle », mobility)
- Additional activities: training, teaching experience, expertise.
- Valorization of results
 - Publications
 - Conferences
 - patents

The objectives of the PhD

Why pursue a PhD after an engineering degree? (1/2)

- Acquire initial professional experience
- Acquire high-level skills in a scientific field
 - Contribution to a research domain
 - Publications of your research results
 - Acquire methodologies in the field of research
- Development of a professional network

The objectives of the PhD

Why pursue a PhD after an engineering degree? (2/2)

- Become a recognized expert in a scientific field
- Obtain an internationally recognized degree
- Training a future teacher/researcher
- Exploring a new research domain
- Valuable initial professional experience for career advancement

Examples of PhD subjects

- LCIS:
 - Co4SYS:
 - « *Commandabilité des systèmes complexes : une approche par la théorie de l'information* »
 - CTSYS:
 - « *Intégration sécurisée d'objets connectés dans les applications pervasives* »
 - ORSYS:
 - « *Étude de nouvelles fonctions radiofréquences à base d'antennes à onde de fuite composite main droite/gauche* »
- Others:
 - « *Authentification par empreinte radio pour l'IoT* »

Before PhD thesis (1/2)

Types of PhD thesis

- Academic PhD thesis, helds in a research laboratory
 - Funding: Doctoral contract, ANR project, Region, ...
 - only one thesis supervisor or one thesis supervisor and one supervisor
- PhD in “Cotutelle” with a foreign university
 - A joint doctoral program is a co-supervised doctoral program that allows students to receive a degree from each of the universities of origin of the two thesis supervisors.
 - It takes place at a French institution and a foreign institution and allows for the awarding of two degrees by each institution.
 - Salary: > 2200 € brut (> 1750 € net)
 - Teaching: In additions of student research's
Salary: 43.5 €/hour (64h/year max)
- CIFRE PhD thesis, held in a company, with a research laboratory (Have a foot in both worlds)
 - Public-private partnership
 - researchProject with the objective of creating value
 - Employment contract with the companyCareer
 - advancement within the company, taking experience into account

Before PhD thesis (2/2)

Preparing your thesis project

- Use your experience to assess your abilities and desire to pursue a PhD
 - Industrial Project
 - PFE: possibility of doing this in a research laboratory
 - Assess your aptitude and desire to solve problems
 - Various projects at ESISAR
- Find a thesis topic
 - Depend on your Cursus (major, minors, ...)
 - Depends of your interests
- Find a
 - « Directeur(s) de thèse » (HdR)
 - « Encadrant(s) de thèse » (PhD)

PhD steps

- Steps of the PhD thesis:
 - State of the art
 - (Formulation of a research question)
 - Theoretical development
 - Experimentations
 - Publications
 - PhD thesis writing
 - PhD defense

Doctoral school

- A doctoral school is an educational structure dedicated for PhD students.
- Inscription must be validate each year
- Mandatory trainings (~100 hours):
 - PhD school trainings (ADUM, ...)
 - Summer/Winter schools (ARTISAN, ...)
- « Comité de Suivi Individuel » (CSI), a committee that evaluate your PhD advancement and its supervision

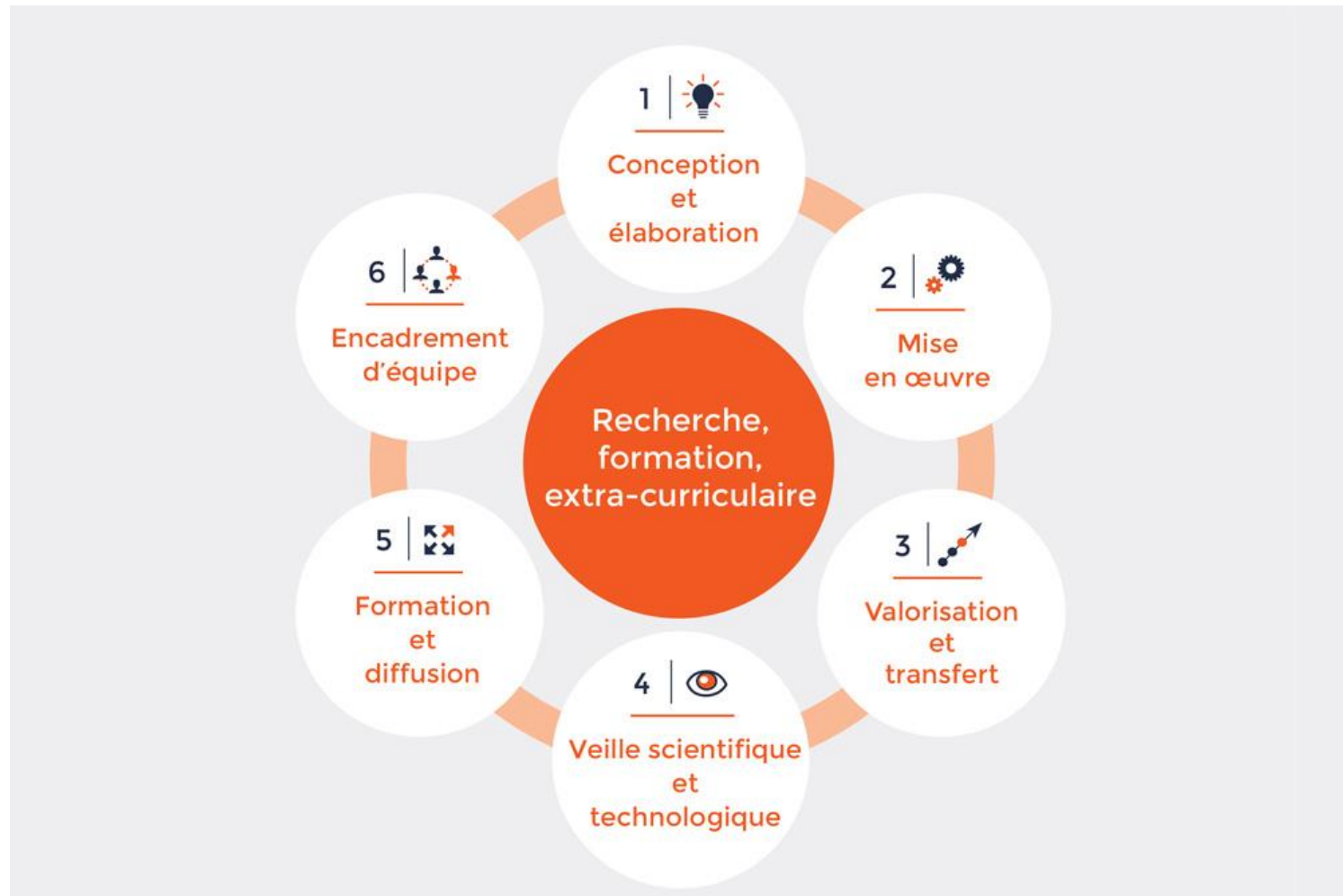
PhD thesis and defense

- Steps (in France) :
 1. The PhD candidate needs to write a PhD manuscript, answering to his research question(s) and presenting his contributions.
 2. The PhD manuscript is reviewed by a committee (HDR, Professor) : « les rapporteurs ».
 3. The PhD candidate does an oral presentation in front of a committee composed of :
 - « Un président de jury »
 - « Des rapporteurs »
 - « Des examinateurs ».

The advantages of a PhD (1/2)

- The advantages of a PhD thesis
 - The scientific research methodology
 - The expertise on a specific domain
 - The capacity to present results
 - Opportunity to pursue an academic career
 - Create a network through conferences
 - International knowledge of the grade
 - Innovation and creativity

The advantages of a PhD (2/2)

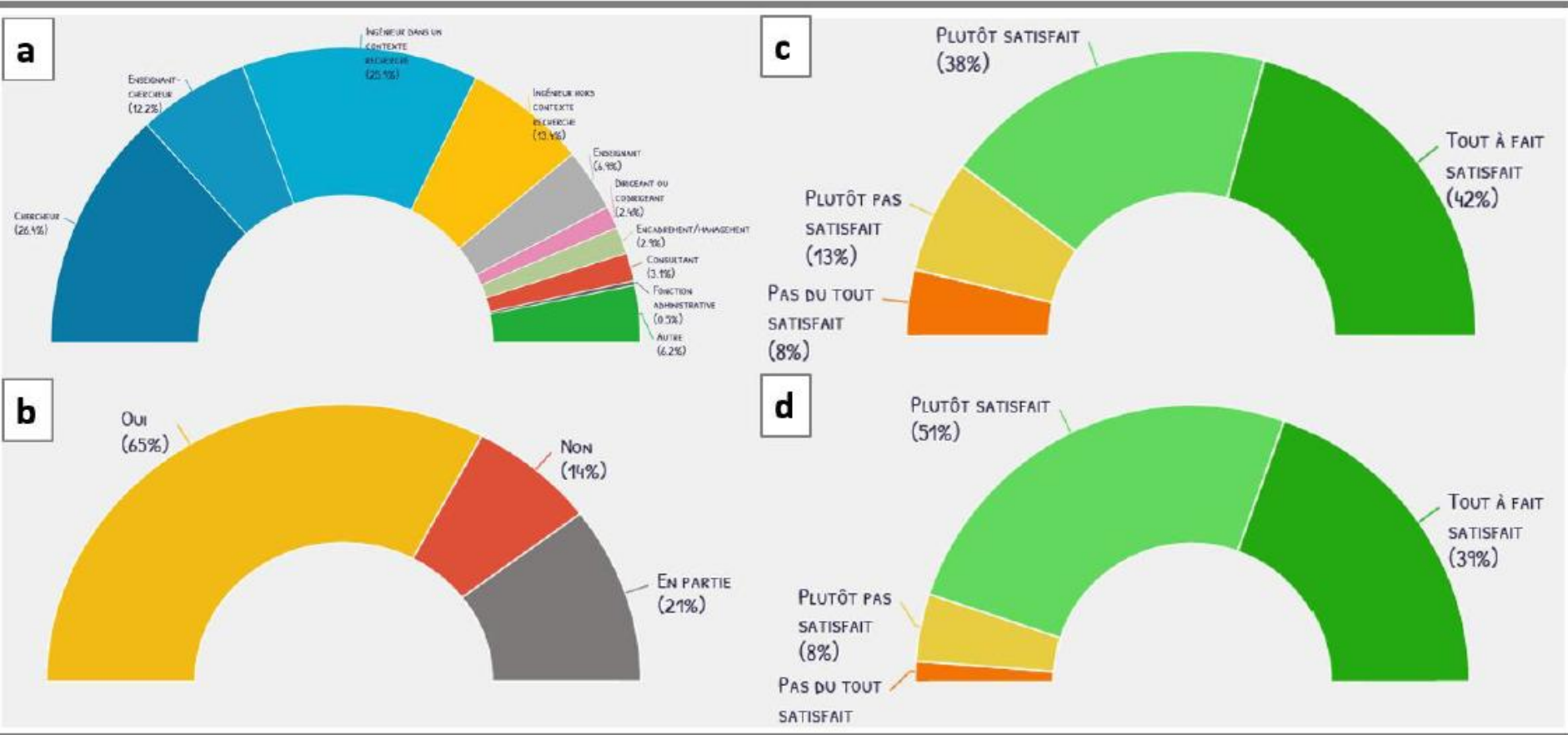


The 6 skill blocks developed during the doctorate defined by the RNCP [1, 2]

Opportunities (1/2)

- The opportunities:
 - Research: Post-doc, researcher, research engineer, ...
 - Teaching: Lecturer, associate professor, ...
 - Institution: ANSI, CEA, DGA, ...
 - Industries: Researcher, engineer, ...

Opportunities (2/2)



UGA Doctoral College Activity Report (2021-2022) [1, 3]
 (a) Actual job (b) Satisfaction in regards to PhD thesis (c) Usefulness of the PhD thesis (d) Satisfaction in regards to the actual job

Testimonials of PhD students (1)

Testimonials of actual and former PhD students (from LCIS and others)

Format:

- Parcours et motivations initiales
- Type de thèse
- Financement
- Sujet (et encadrement)
- École doctorale
- Déroulement
- (Avantages et débouchés): Pour les anciens

Testimonials of PhD students

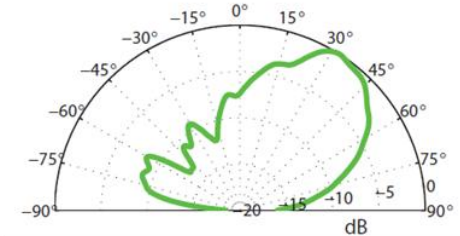
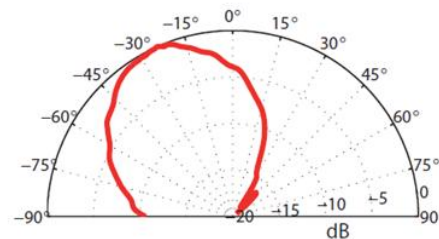
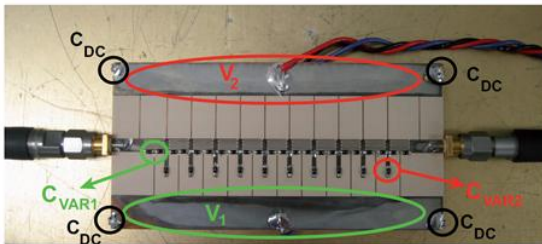
Romain Siragusa

Curriculum Vitae (CV)

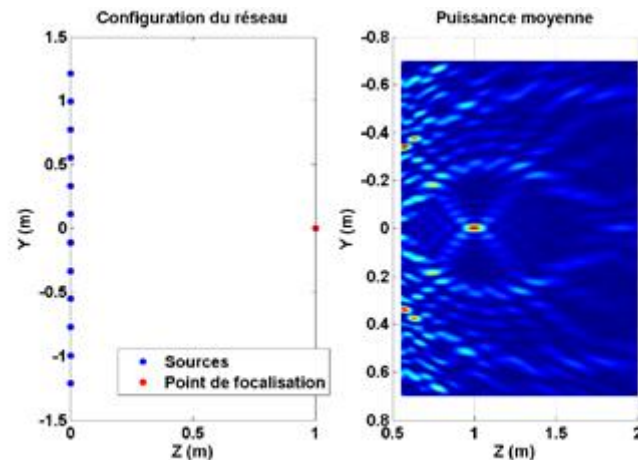
- 2001-2006 : Engineering degree at Grenoble INP – Esisar
- 2006 : Master degree in optical and radiofrequency at Grenoble INP
- 2007-2009 : PhD in optical and radiofrequency at LCIS
 - Funded by ministry grant with EEATS doctoral school
 - Teaching at Esisar 64 hours/year
 - Collaboration with Christophe Caloz from Polytechnique Montréal
 - Title: « Étude de nouvelles fonctions radiofréquences à base d'antennes à onde de fuite composite main droite/gauche »
- 2009-2010 : Temporary teaching assistant at Esisar
- 2010-2011 : Researcher at CEA LETI in Grenoble
 - Subject : Design of millimeter wave antennas embedded on silicon chip
- 2011-Now : Associate Professor in Grenoble INP Esisar
 - Researcher in the ORSYS team

PhD contributions

- Axis 1 : Automated design and models of Composite Right/Left handed antennas (CRLH LWA)
- Axis 2 : Efficient electronically scanning CRLH LWA



- Axis 3 : RF focalisation using phased array antennas



Pros and Cons

Pros

Technical expertise

International context :
collaboration, conferences

Work in autonomy

Cons

Salary during the phd compare
to private positions

Work in autonomy

job opportunities in the public
sector are uncertain

Testimonials of PhD students

Arthur Desuert

Curriculum Vitae (CV)

My journey (in brief)

2017 – 2020: Engineering degree at Grenoble INP – Esisar

Computer science, networks and cybersecurity

Mainly software courses, with a touch of hardware



2020 – 2023: Ph.D. in computer science at LCIS

Funded by Chaire TRUST, from Fondation Grenoble INP

With MSTII doctoral school

Wireless protocols and hardware security in IoT

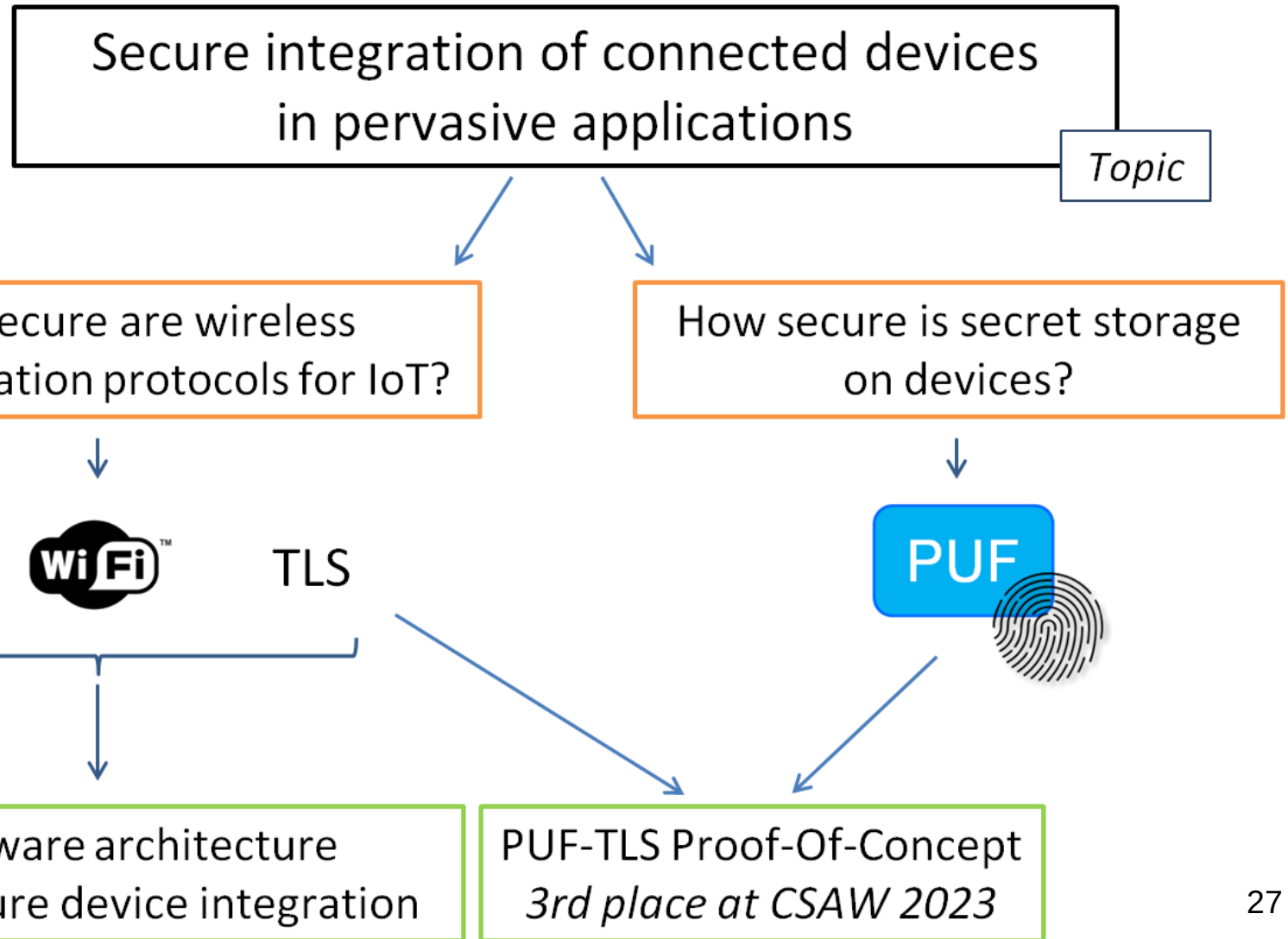


2024 – now: Research engineer at LCIS

Head of support services

Scientific support and research data management

My Ph.D. – topic and contributions



My Ph.D. – Pros and Cons

(From my subjective point of view)

Pros

- Great liberty of action
- International scope
- Technical expertise AND transversal skills

Cons

- Captain of you own boat
- Salary (often) not attractive compared to private positions
- Lots of writting

My Ph.D. – Pros and Cons

(From my subjective point of view)

Pros

- Great liberty of action
- International scope
- Technical expertise AND transversal skills
- You can make people call you “Doctor”

Cons

- Captain of you own boat
- Salary (often) not attractive compared to private positions
- Lots of writting

Testimonials of PhD students

Pierre-Alain Toupance

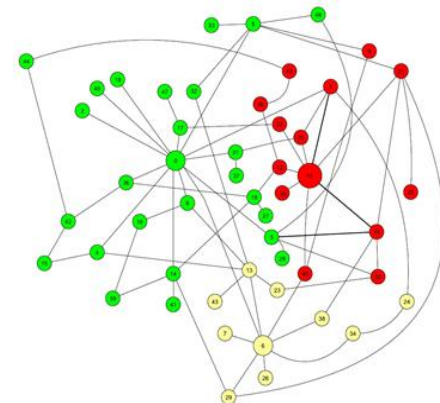
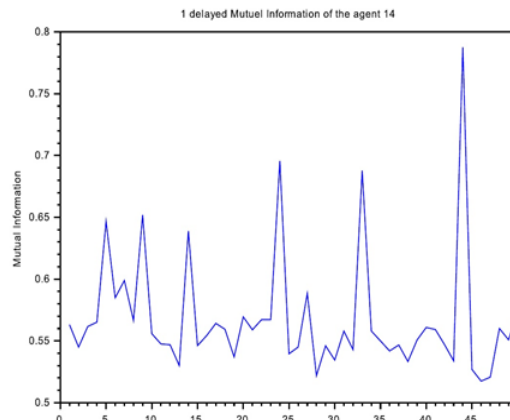
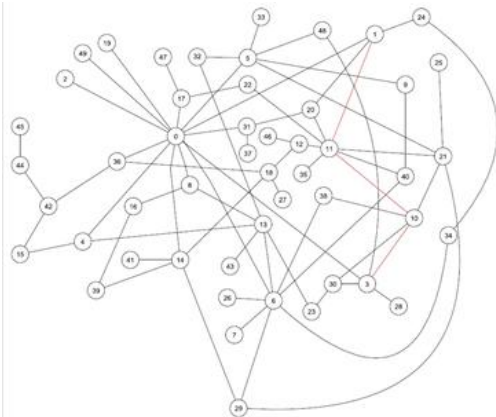
Pierre-Alain Toupance - CV

- 1994 : Agregation of mathematics
- 1994 -1999 : Teacher of mathematics in secondary schools (Cairo, Montreal, La Réunion)
- 2000 : Master degree in applied mathematics in Lyon 1
- 2000 - 2010 : PRAG of Mathematics in the University Lyon1
- 2010 - now : PRAG of Mathematics in Grenoble – INP ESISAR
- 2018 - 2024 : PHD in Automatic and applied mathematics
 - Cotutelle : UGA and University of Geneva
 - Title: *Controlability of complex system* : an information theoretic approach.

Objectives of the thesis

Use Claude Shannon's information theory to study problems of controllability in complex systems whose topology is unknown.

- Determine the topology of the system.
- Detect changes in topology over time.
- Evaluate how influence spreads.
- Identify the most influential elements of the system in order to facilitate control of the system (i.e., bring the system to a desired state).
- Build communities (clusters) within the system in order to construct a simplified model.



Pros and Cons

Pros

Acquire scientific and technical skills in many fields.

Presenting results at conferences or publishing articles in journals.

The pleasure of having an intellectual activity with a certain degree of freedom.

Cons

Often working alone (due to my particular circumstances).

Difficulty managing the thesis with everything else.

Report writing.

Testimonials of PhD students

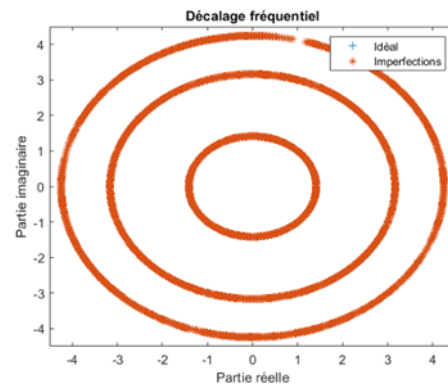
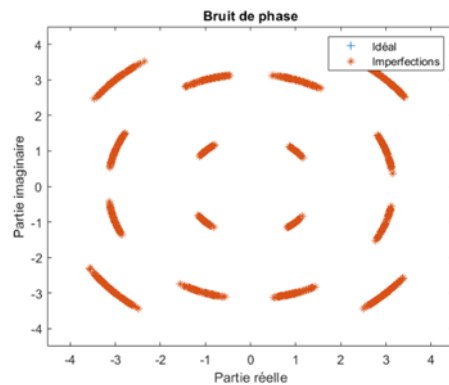
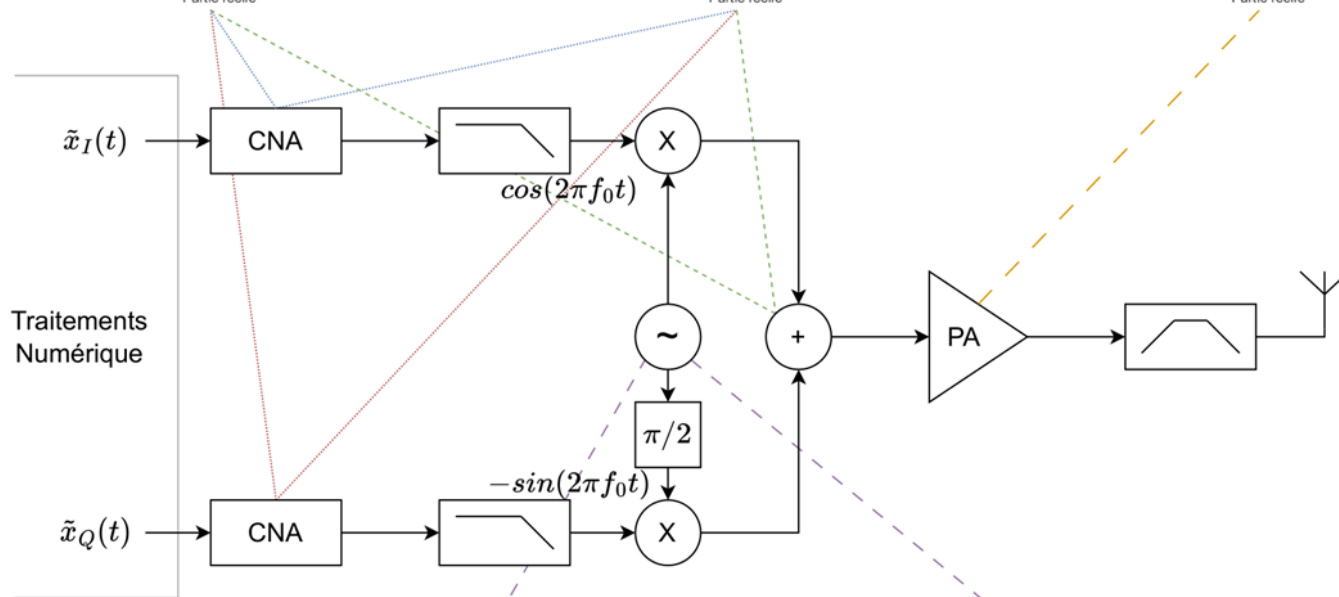
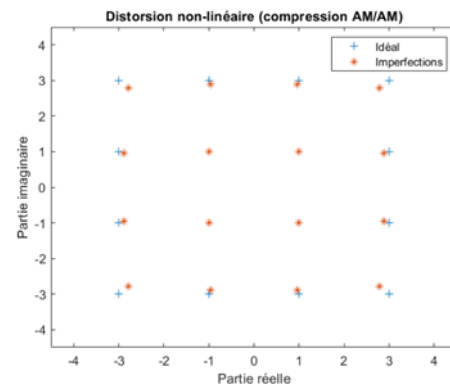
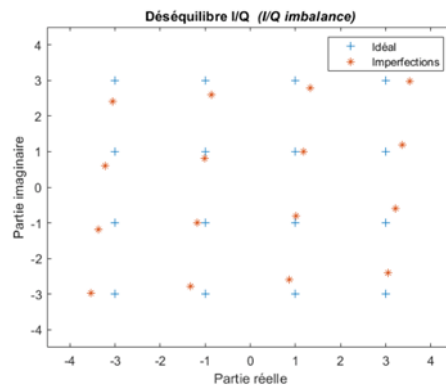
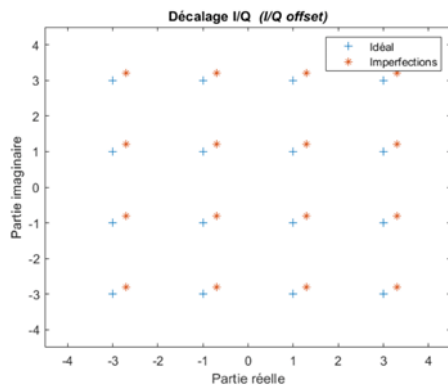
Louis Morge-Rollet

Curriculum Vitae (CV)

- 2014-2016 : Preparatory cycle at Grenoble INP – Esisar
- 2016-2019 : Engineering degree at Grenoble INP – Esisar
 - EIS : « Electronique et Informatique des Systèmes Embarqués »
 - ESE (4A) : « Électronique des Systèmes Embarqués »
 - SCN (5A) : Systèmes de Communication Numériques
- 2019-2023: PhD in Telecommunications at Lab-STICC (ENSTA Bretagne)
 - Funded by Army ministry (« chaire DGA IoT » / ENSTA Bretagne)
 - Doctoral school : MathSTIC, then SPIN
 - Teaching at ENSTA Bretagne : 64 hours/year
 - Title: « Authentification par empreinte radio pour l'IoT »
- 2022-2024 : Lecturer at ENSTA Bretagne
- 2024-Now : Associate Professor in Grenoble INP - Esisar
 - Researcher in the ORSYS team

PhD subject and contributions

- Goal : Authentify a particular Radio Frequency (RF) transmitter, by leveraging on its impairments (RFF : RF Fingerprint), in Internet of Things (IoT) context
- Research question : « Quelles sont les propriétés nécessaires à l'utilisation du RFF dans un contexte IoT et quelles sont les méthodes (et leurs implémentations) qui en découlent ? »
- Contributions :
 1. Properties of RFF methods used in IoT context : Scalability, adaptability and complexity
 2. Method 1 : Siamese network on I/Q signals for RF Fingerprinting
 3. Method 2 : RF eigenfingerprints
 4. Implementations and characterization (latency, memory size, ...) of the proposed methods on embedded targets (SoC, FPGA)



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Pros and Cons

Pros

The liberty to explore several ways to solve a problem

Studying a cutting-edge field and creating new knowledges

Collaboration with other PhD students, researchers, ...

Cons

The pressure about the publications

The lack of real-world cases/data

Writing the PhD manuscript can be (really) long

Bibliography

- [1] Le doctorat à l'Université Grenoble Alpes – Collège Doctoral UGA
- [2] [Les compétences du doctorat – Collège Doctoral UGA](#)
- [3] [Rapport d'activité 2021-2022 – Collège Doctoral UGA](#)